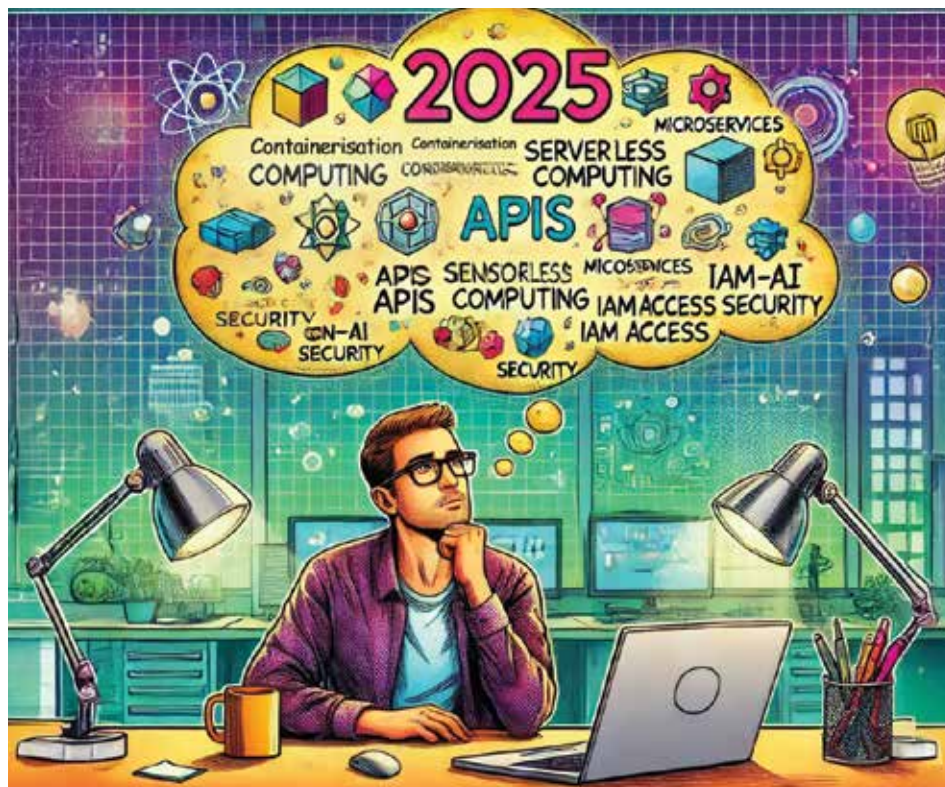


Top 7 Open Source Technologies to Learn and Master in 2025

Open source technologies like containerisation, microservices with serverless computing, generative AI, API-driven development, robust IAM systems, observability, and security are defining the future of software development. If you are entering the job market this year, you will do well to master these.



The rapid advancement of technology continues to shape industries worldwide, and open source solutions are leading this transformation. For college students pursuing a career in software development, mastering open source technologies offers a gateway to exciting job opportunities and innovative projects. This guide highlights seven critical areas that developers should prioritise in 2025: containerisation, microservices with serverless computing, generative AI (GenAI), API-driven development, robust identity and access management

(IAM) systems, observability, and security. By gaining expertise in these domains, you'll be equipped to tackle modern challenges and create scalable, impactful solutions.

Containerisation: Simplifying application deployment

Containerisation has revolutionised how applications are developed, deployed, and maintained. By bundling software and its dependencies into lightweight, portable containers, developers ensure consistency across different environments. Technologies like Docker and Kubernetes have become industry

standards, making containerisation a must-learn skill. Docker simplifies container creation, sharing, and deployment, while Kubernetes manages and orchestrates containers at scale. Podman offers a daemonless alternative for container management, providing flexibility in diverse setups.

Containerisation is vital in 2025 because it supports scalability by integrating seamlessly with microservices, optimises resource usage for cost efficiency, and has widespread adoption, making it a highly sought-after skill. To get started, developers can follow Docker tutorials